RAM-Forensics: Detecting ACPI Rootkits

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> whoami

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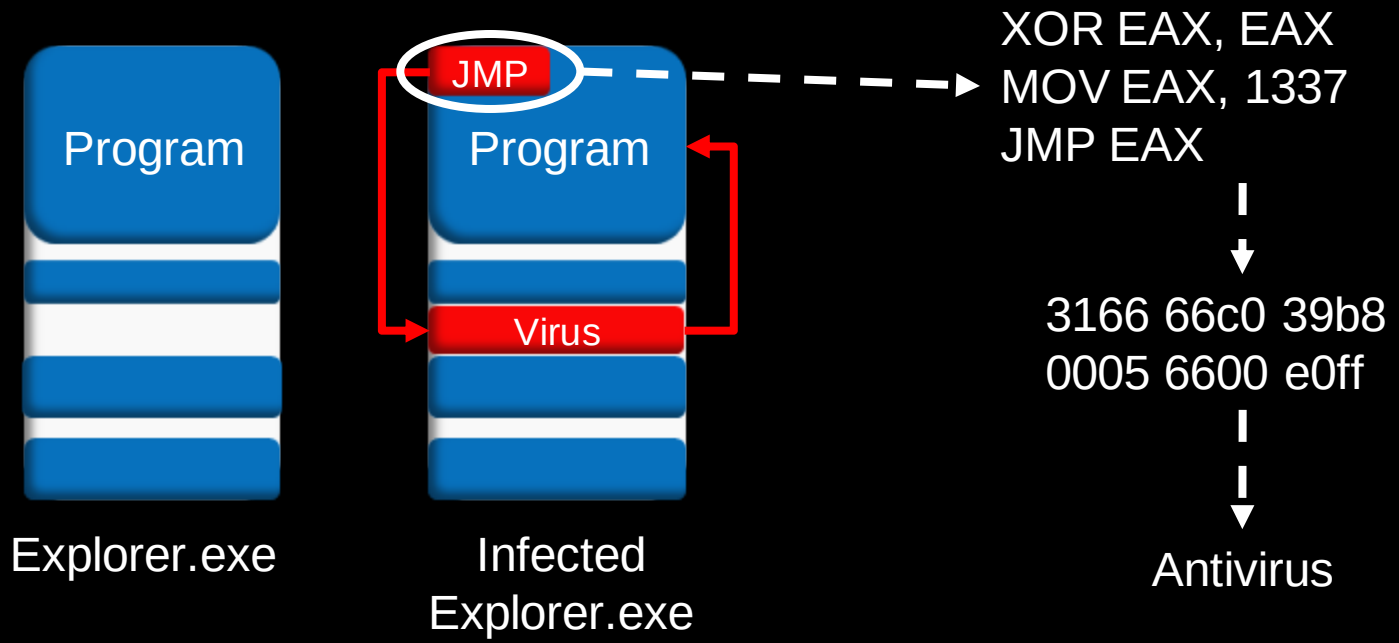
PhD: „Malware tolerance – Distributing trust over multiple devices“

Experience:

- SOC Analyst
- IT-Security Consultant (building SOCs, DFIR, Pentests)
- Incident Responder @Airbus CyberSecurity

More Blue-Teamer

File-based Malware



File-less Malware

- File-less = no files
- File-less ≠ RAM-only

- often: Powershell
- Frameworks: Powershell-Empire, PowerSploit

Process Hollowing

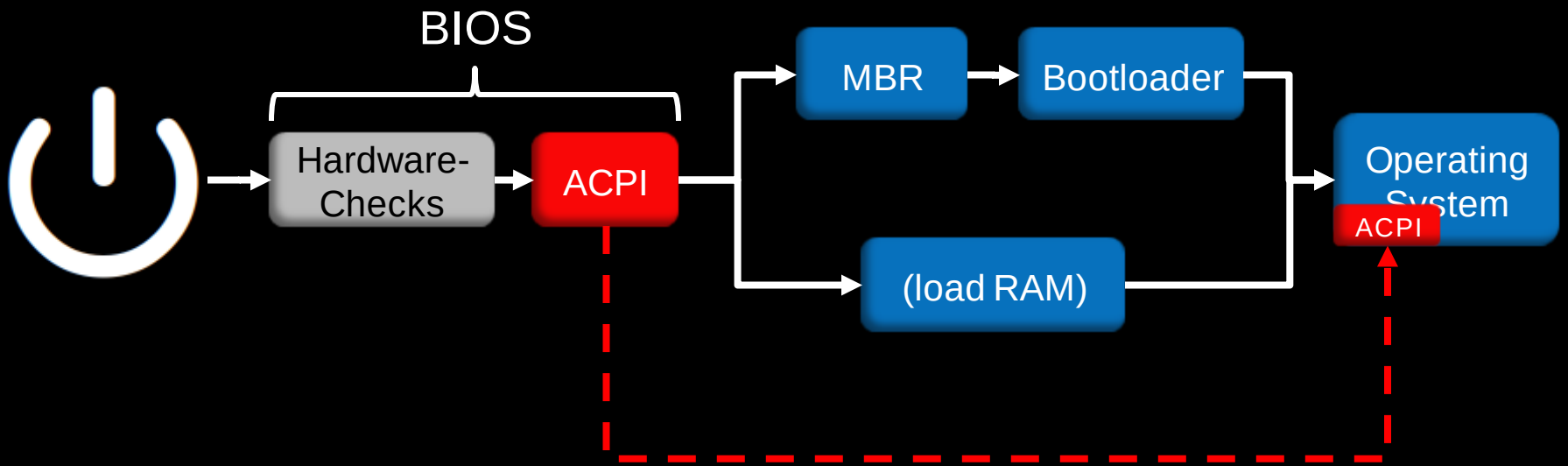
1. Start a copy of legitimate process
2. Suspend process
3. Deallocate (unmap) and replace code
4. Resume process

→ Looks legitimate: name, path, commandline (e.g. Stuxnet)

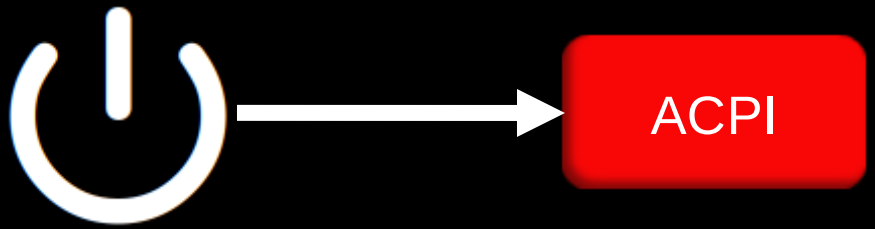
Rootkits

- root + toolkit = software for longterm root/admin access
- Focus: Stealth
- Different levels: user-mode rootkit, kernel rootkit, etc.
- Often paired with backdoor

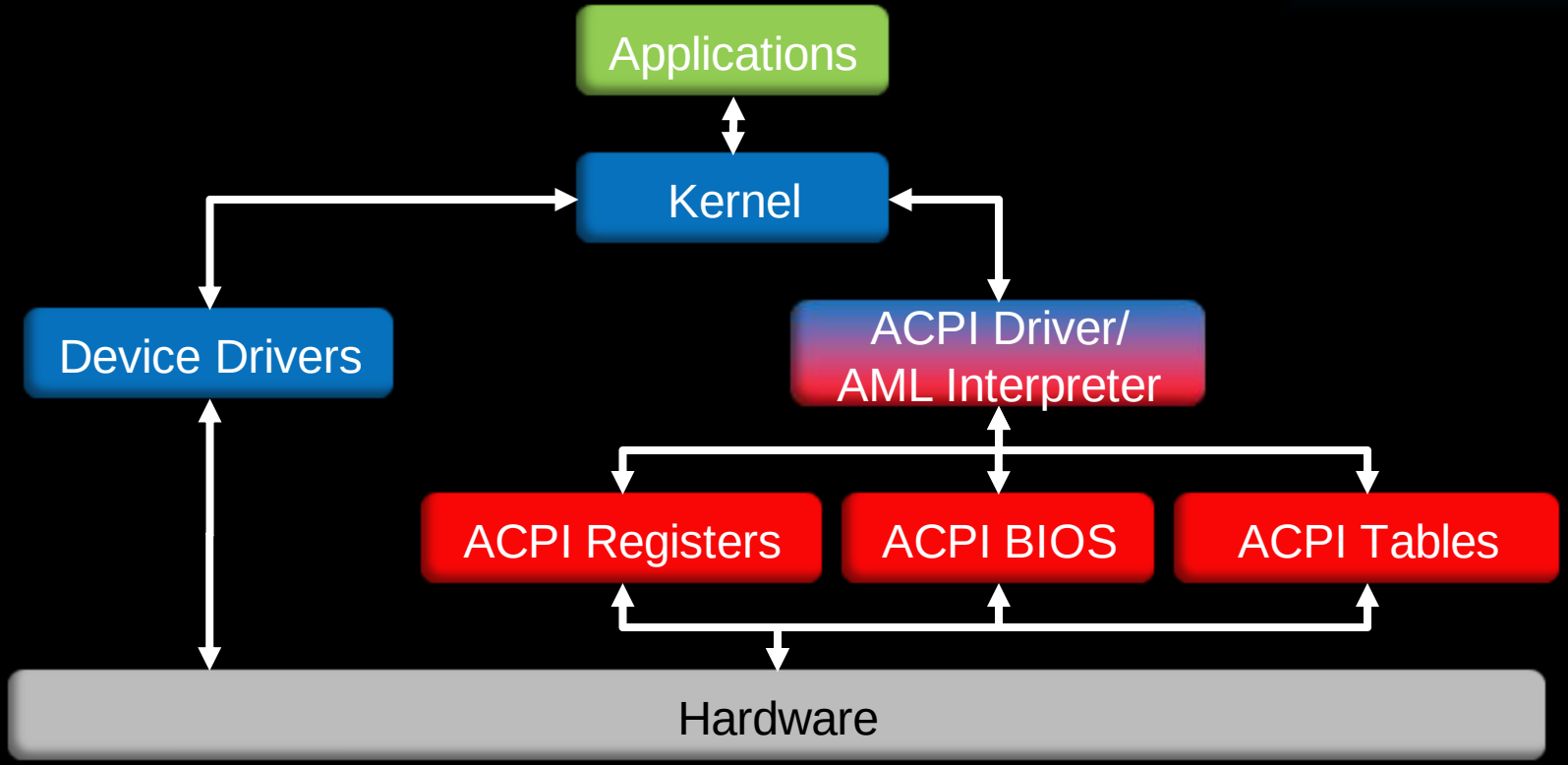
Advanced Configuration and Power Interface (ACPI)



ACPI



ACPI Components



ACPI Tables

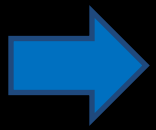
- RSDP „RSD PTR“
  - RSDT / XSDT
    - FADT
      - DSDT (cannot be unloaded)
      - FACS
    - (optional) SSDT
    - (optional) OEMx
    - … (many more)

ACPI Code: AML/ASL

DSDT.aml

```
00001510: 4241 5430 085f 4849 440c 41d0 0c0a 085f  BAT0._HID.A...._
00001520: 5549 4400 101f 5c5f 4750 4514 185f 4c30  UID...\_GPE.._L0
00001530: 3000 865c 2f03 5f53 425f 5043 4930 4241  0..\/._SB_PCI0BA
00001540: 5430 0a81 5b80 4342 4154 010b 4040 0a08  T0..[.CBAT..@@..
00001550: 5b81 1043 4241 5403 4944 5830 2044 4154  [..CBAT.IDX0 DAT
00001560: 3020 5b86 4605 4944 5830 4441 5430 0353  0 [.F.IDX0DAT0.S
```

ACPI Machine Language (AML)



DSDT.dsl

```
Device (BAT0)
{
    Name (_HID, EisaId ("PNP0C0A"))
    Name (_UID, Zero)
    Scope (\_GPE)
    {
        Method (_L00, 0, NotSerialized)
        {
            Notify (\_SB.PCI0.BAT0, 0x81)
        }
    }

    OperationRegion (CBAT, SystemIO, 0x4040, 0x08)
    Field (CBAT, DWordAcc, NoLock, Preserve)
    {
        IDX0,   32,
        DAT0,   32
    }
```

Disassembled ACPI Source Language (DSL)

ACPI Rootkit

- John Heasman 2006 [1]
- Idea:
  - OperationRegion + SystemMemory + Store
  - Use functions being called at runtime
  - e.g. temperature control

ACPI Rootkit

- PoC: Overwrite unused systemcall (`sys_ni_syscall`) to elevate privileges

```
OperationRegion(NISC, SystemMemory, 0x12BAE0, 0x40)
Field(NISC, AnyAcc, NoLock, Preserve)
{
  NICD, 0x40
}
Store(Buffer () {0xFF, 0xD3, 0xC3, 0x90, 0x90, 0x90, 0x90,0x90}, NICD)
```

{ call ebx; retn; nop; nop; nop; nop; nop }

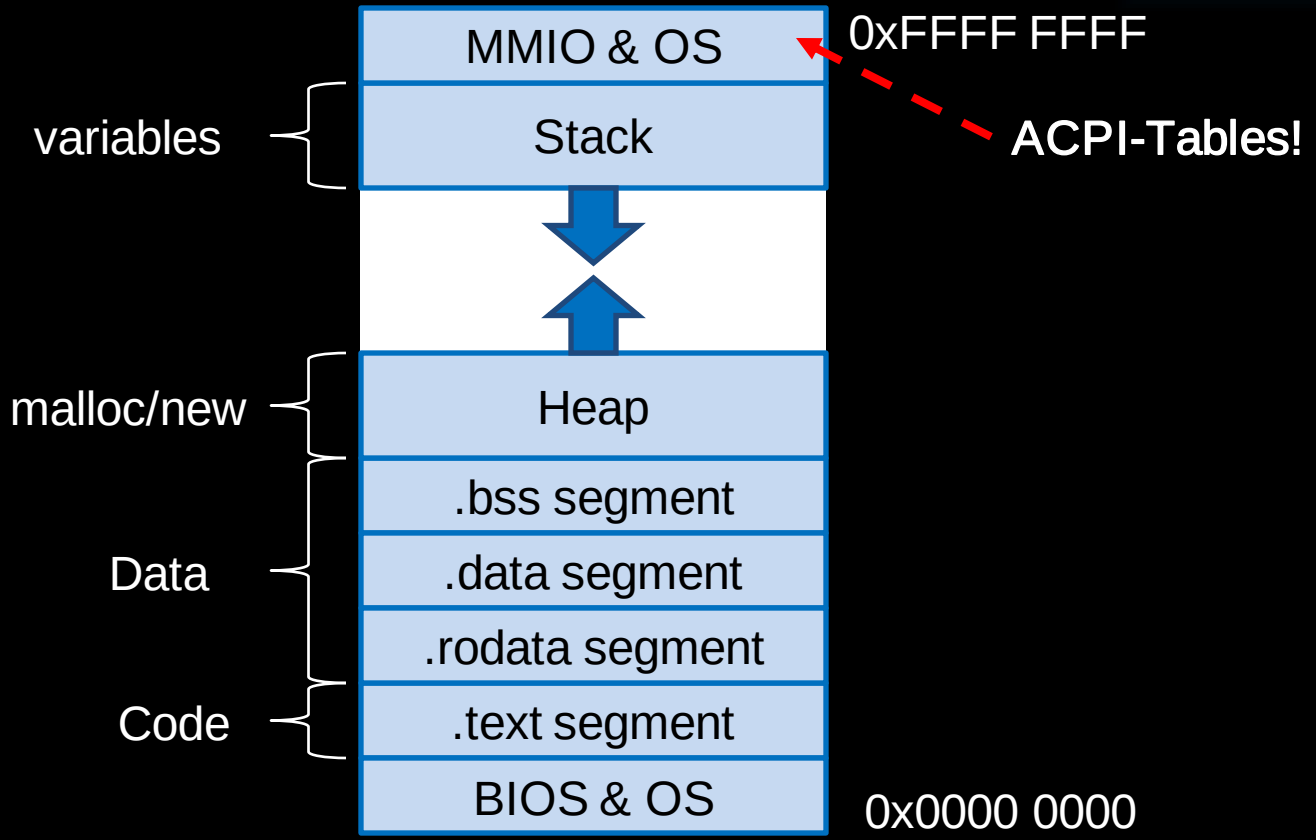
Source [1]

- `sys_ni_syscall(evilfunction);`

RAM Forensics

- Running processes: RAM → Cache → CPU
- Operating System structures (e.g. _EPROCESS list)
- Forensics: lie-detection („screen“ vs HDD/RAM)

RAM Basics


MMIO & OS
0xFFFF FFFF
variables
Stack
ACPI-Tables!
malloc/new
Heap
.bss segment
Data
.data segment
.rodata segment
Code
.text segment
BIOS & OS
0x0000 0000

volatility

- by: Michael Cohen
- open-source RAM analysis framework
- commandline only
- Volatility2 (python2) vs Volatility3 („beta“, python3)

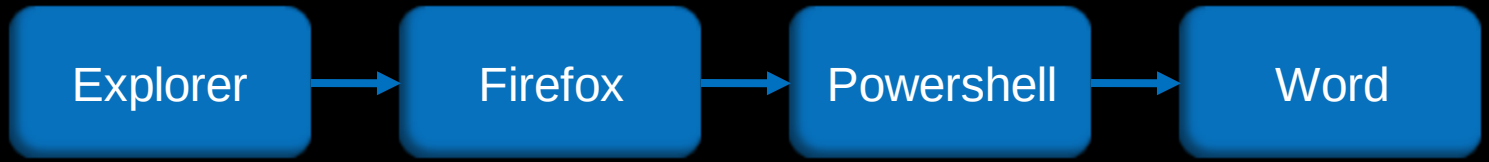
volatility Basics

```
$ python vol.py -f ~/Desktop/win7_trial_64bit.raw --profile=Win7SP0x64 pslist
Volatility Foundation Volatility Framework 2.4
Offset(V)          Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exi
------------------ -------------------- ------ ------ ------ -------- ------ ------ ------------------- ---
0xfffffa80004b09e0 System                    4      0     78      489 ------      0 2012-02-22 19:58:20
0xfffffa8000ce97f0 smss.exe                208      4      2       29 ------      0 2012-02-22 19:58:20
0xfffffa8000c006c0 csrss.exe               296    288      9      385      0      0 2012-02-22 19:58:24
```

Source [2]

volatility Basics

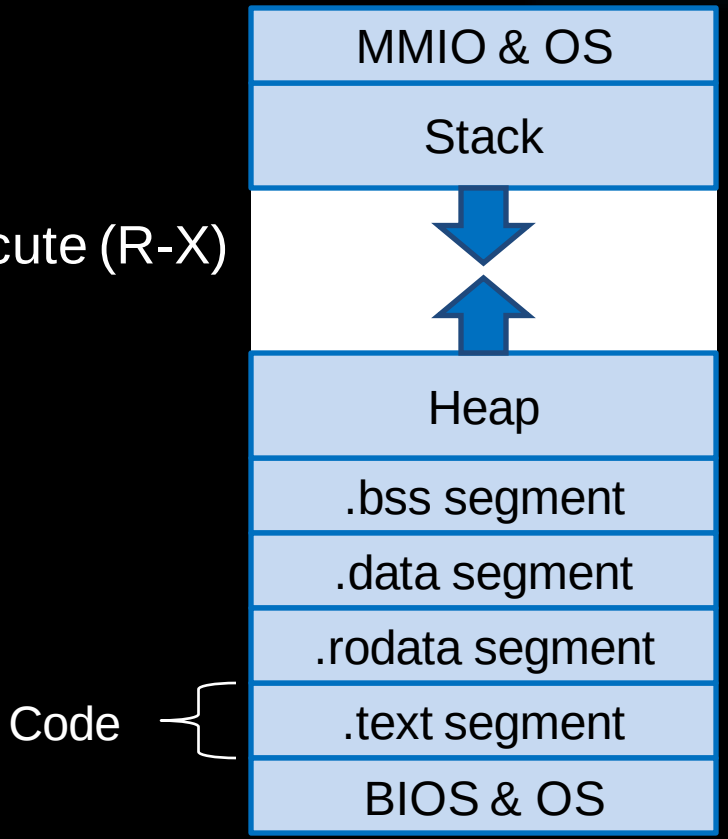
_EPROCESS list of operating system:



Advanced volatility

Fileless malware:

- `volatility malfind`
- .text segement read-only and execute (R-X)
- Injection → write-able (-W-)
- Malfind: find code-pages with WX



Advanced volatility

Process Hollowing:

- `Suspend + Unmap + Replace + Start`
- `volatility ldrmodules`

```
$ python vol.py -f ~/Desktop/win7_trial_64bit.raw --profile=Win7SP0x64 ldrmodules
Volatility Foundation Volatility Framework 2.4
Pid      Process              Base               InLoad InInit InMem MappedPath
-------- -------------------- ------------------ ------ ------ ----- ----------
     208 smss.exe             0x0000000047a90000 True   False  True  \Windows\System32\smss.exe
     296 csrss.exe            0x0000000049700000 True   False  True  \Windows\System32\csrss.exe
     344 csrss.exe            0x0000000000390000 False  False  False \Windows\Fonts\vgasys.fon
     344 csrss.exe            0x00000000007a0000 False  False  False \Windows\Fonts\vgaoem.fon
```

Source [2]

Advanced volatility

Process Hollowing:

- Suspend + Unmap + Replace + Start
- volatility ldrmodules

```
$ python vol.py -f ~/Desktop/win7_trial_64bit.raw --profile=Win7SP0x64 ldrmodules
Volatility Foundation Volatility Framework 2.4
Pid      Process              Base               InLoad InInit InMem MappedPath
-------- -------------------- ------------------ ------ ------ ----- ----------
     208 smss.exe             0x0000000047a90000 True   False  True  \Windows\System32\smss.exe
     296 csrss.exe            0x0000000049700000 True   False  True  \Windows\System32\csrss.exe
     344 csrss.exe            0x0000000000390000 False  False  False \Windows\Fonts\vgasys.fon
     344 csrss.exe            0x00000000007a0000 False  False  False \Windows\Fonts\vgaoem.fon
```

Source [2]

volatility

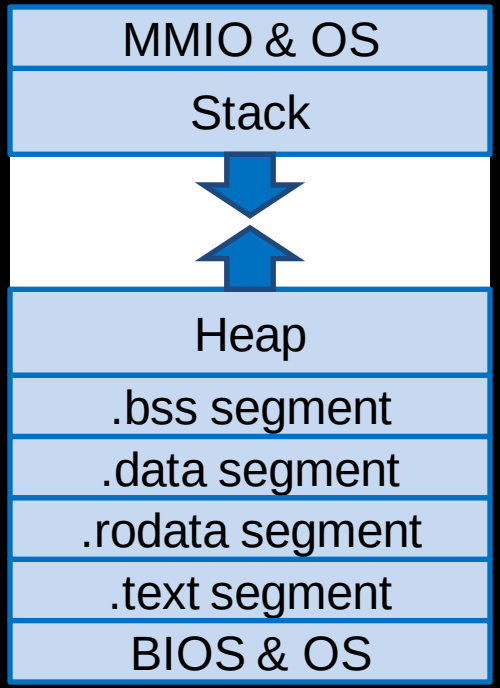
Live-Demo

Advanced volatility

- Fileless Malware: volatility malfind
- Process Hollowing: volatility ldrmodules
- ACPI rootkit: ?

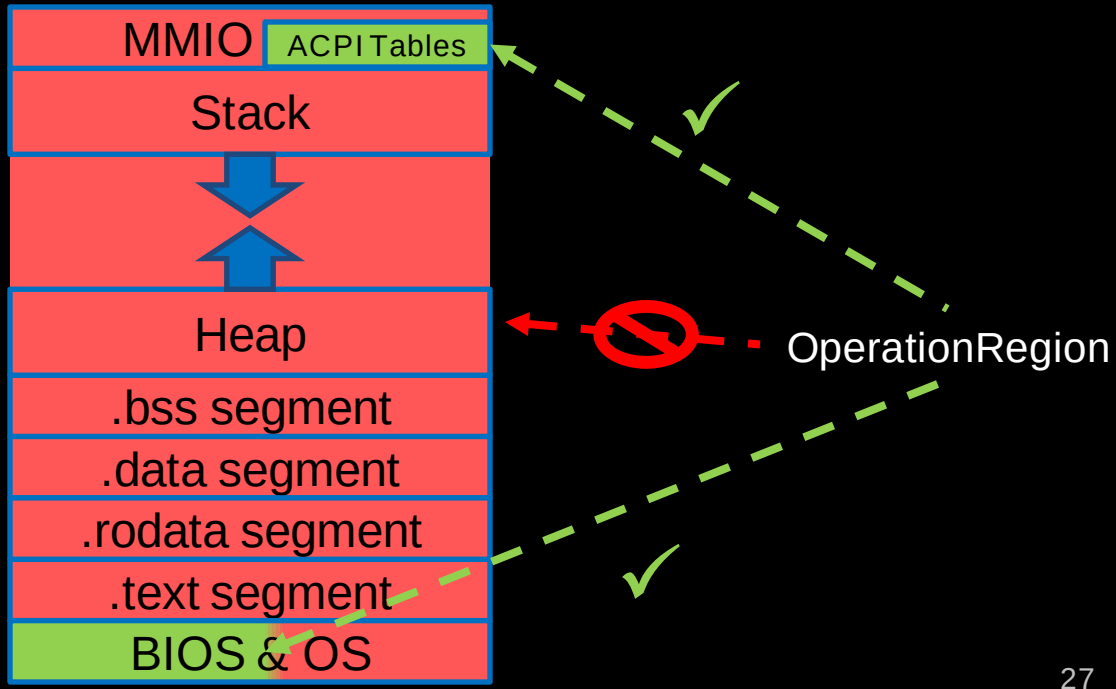
ACPI Rootkit Detection

- What does ACPI not know normally?



ACPI Rootkit Detection

- What does ACPI not know normally?



ACPI Rootkit Detection Tool

```
ctf@localhost ~/p/ACPI-rootkit-scan (master)> python2 ~/progs/volatility/vol.py --plugins=. -f ../../Malware/cridex.vmem --profile=WinXPSP2x86 scanacpitables --dump
Volatility Foundation Volatility Framework 2.6.1

table column "Rootkit?" may have values (seems ok/unknown/suspicious/CRITICAL)
File                 Function                                                                  Rootkit?
-------------------- ------------------------------------------------------------------------- -----------
WARNING : volatility.debug    : FADT.X_FirmwareControl and FADT.FirmwareControl set! According to ACPI documentation this is not allowed! - Trying FirmwareControl
0x000f6b50/DSDT.dsl  OperationRegion (OEMD, SystemMemory, 0x1FEFFE89, 0x00000034)              seems ok
0x000f6b50/DSDT.dsl  OperationRegion (REGS, PCI_Config, 0x50, 0x30)                            seems ok
0x000f6b50/DSDT.dsl  OperationRegion (RE00, PCI_Config, 0xD8, 0x04)                            seems ok
0x000f6b50/DSDT.dsl  OperationRegion (GEN, PCI_Config, 0xB0, 0x04)                             seems ok
0x000f6b50/DSDT.dsl  OperationRegion (PIRX, PCI_Config, 0x60, 0x04)                            seems ok
0x000f6b50/DSDT.dsl  OperationRegion (PCI, PCI_Config, 0x40, 0x60)                             seems ok
0x000f6b50/DSDT.dsl  OperationRegion (MREG, SystemMemory, MBAS (Arg0), 0x10)                   suspicious
0x000f6b50/DSDT.dsl  OperationRegion (EICH, SystemMemory, (ECFG + 0x4000), 0x4000)             suspicious
0x000f6b50/DSDT.dsl  OperationRegion (SMI0, SystemIO, 0x00000CF0, 0x00000002)                  seems ok
0x000f6b50/DSDT.dsl  OperationRegion (SMI1, SystemMemory, 0x1FEFFEBD, 0x00000090)              seems ok
0x000f6b50/DSDT.dsl  OperationRegion (EREG, SystemMemory, ECFG, 0x4000)                        seems ok
0x000f6b50/DSDT.dsl  OperationRegion (CREG, SystemMemory, Local1, 0x10)                        unknown
0x000f6b50/DSDT.dsl  OperationRegion (CREG, SystemMemory, Local1, 0x01)                        unknown
0x000f6b50/DSDT.dsl  OperationRegion (RE01, PCI_Config, 0x40, 0x04)                            seems ok
0x000f6b50/DSDT.dsl  OperationRegion (RE02, PCI_Config, 0xC4, 0x04)                            seems ok
0x000f6b50/DSDT.dsl  OperationRegion (REGS, PCI_Config, 0x00, 0x04)                            seems ok
0x000f6b50/DSDT.dsl  OperationRegion (AWK0, SystemMemory, (ECFG + 0x0232), 0x20)               suspicious
0x000f6b50/DSDT.dsl  OperationRegion (QREG, SystemMemory, (ECFG + 0x0100), 0x10)               suspicious
0x000f6b50/DSDT.dsl  OperationRegion (LPCS, SystemMemory, ECFG, 0x0400)                        seems ok
```

ACPI rootkit scanner

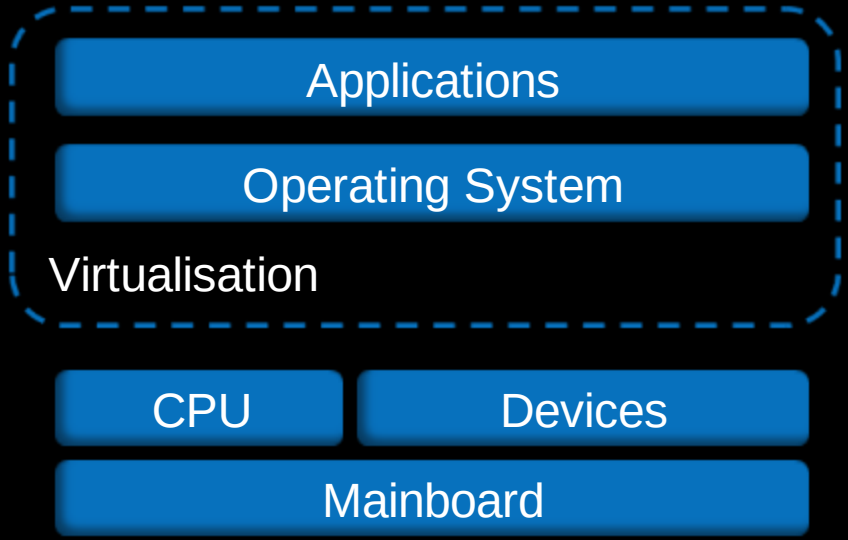
Live-Demo

Removing ACPI Rootkit

Source [6]

Source [7]

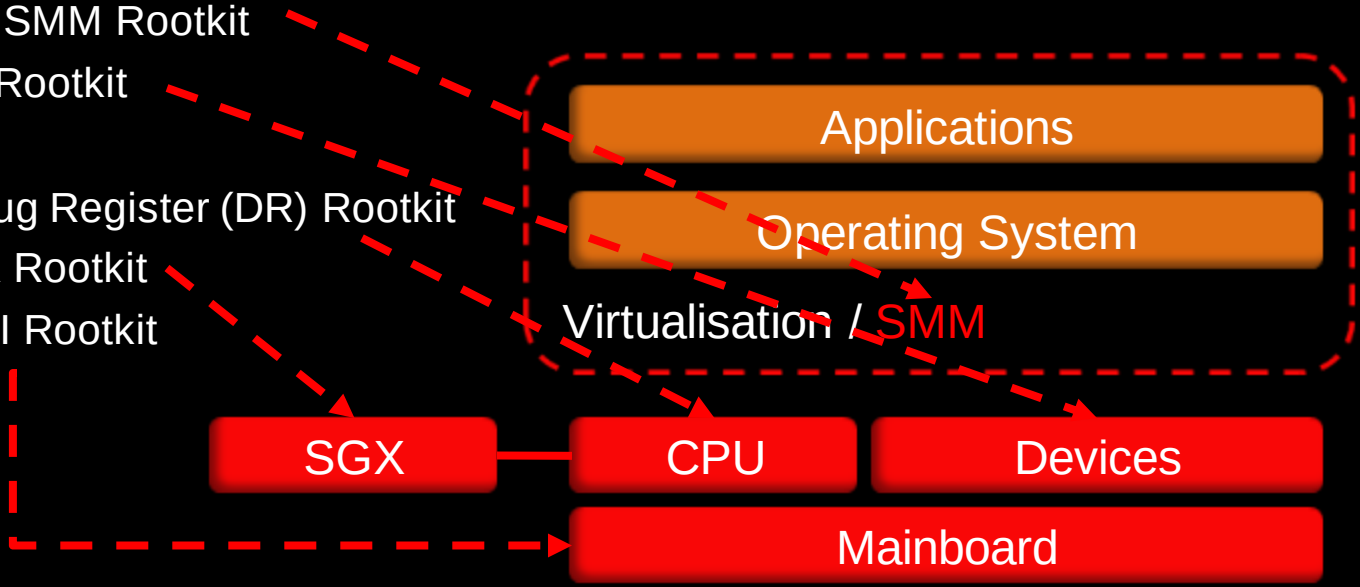
Advanced (Firmware) Rootkits



Advanced (Firmware) Rootkits

1. Intel SMM Rootkit
2. PCI Rootkit

3. Debug Register (DR) Rootkit
4. SGX Rootkit
5. ACPI Rootkit



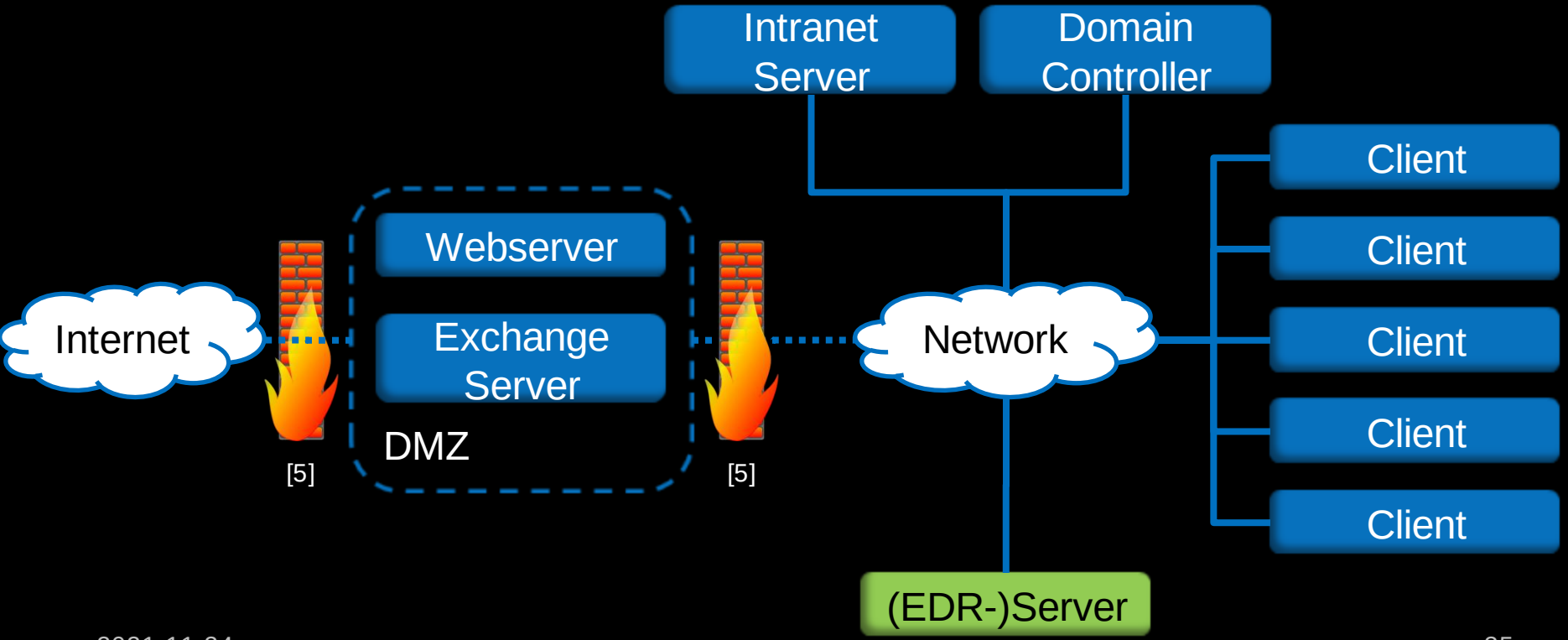
Future: Live-Forensics

- How to take a RAM-image (spontaneously)?
  - specialized software (preinstalled?)
- How servercluster 100 server?

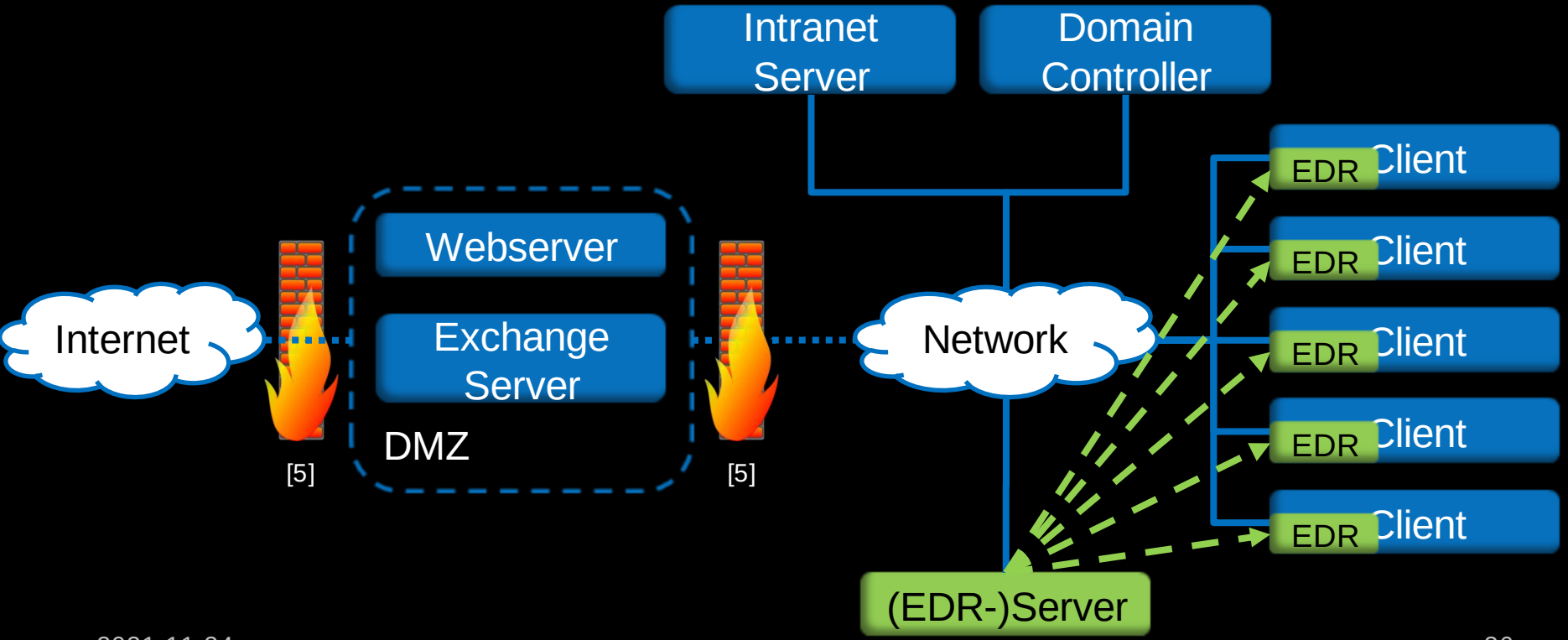
Endpoint Detection & Response (EDR)

- Digital Forensics
- Incident Response
- Client (Agent) & Server

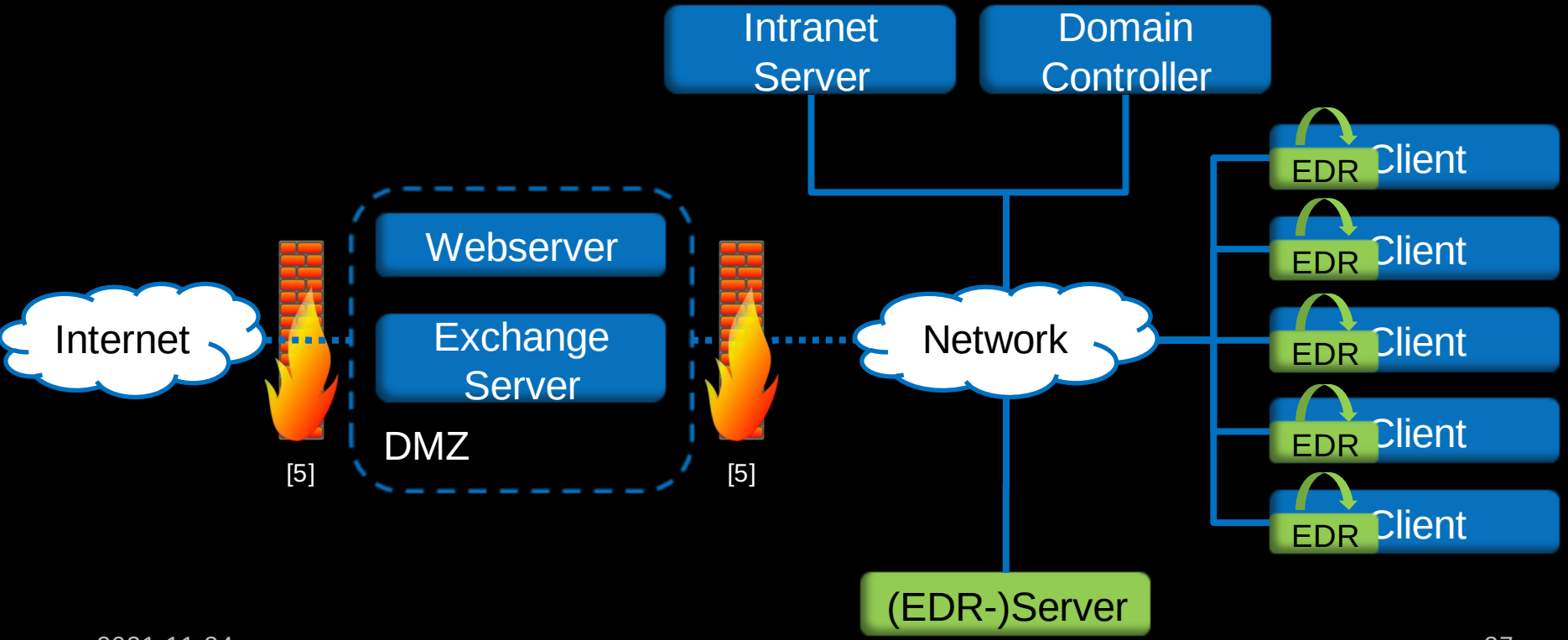
Endpoint Detection & Response (EDR)



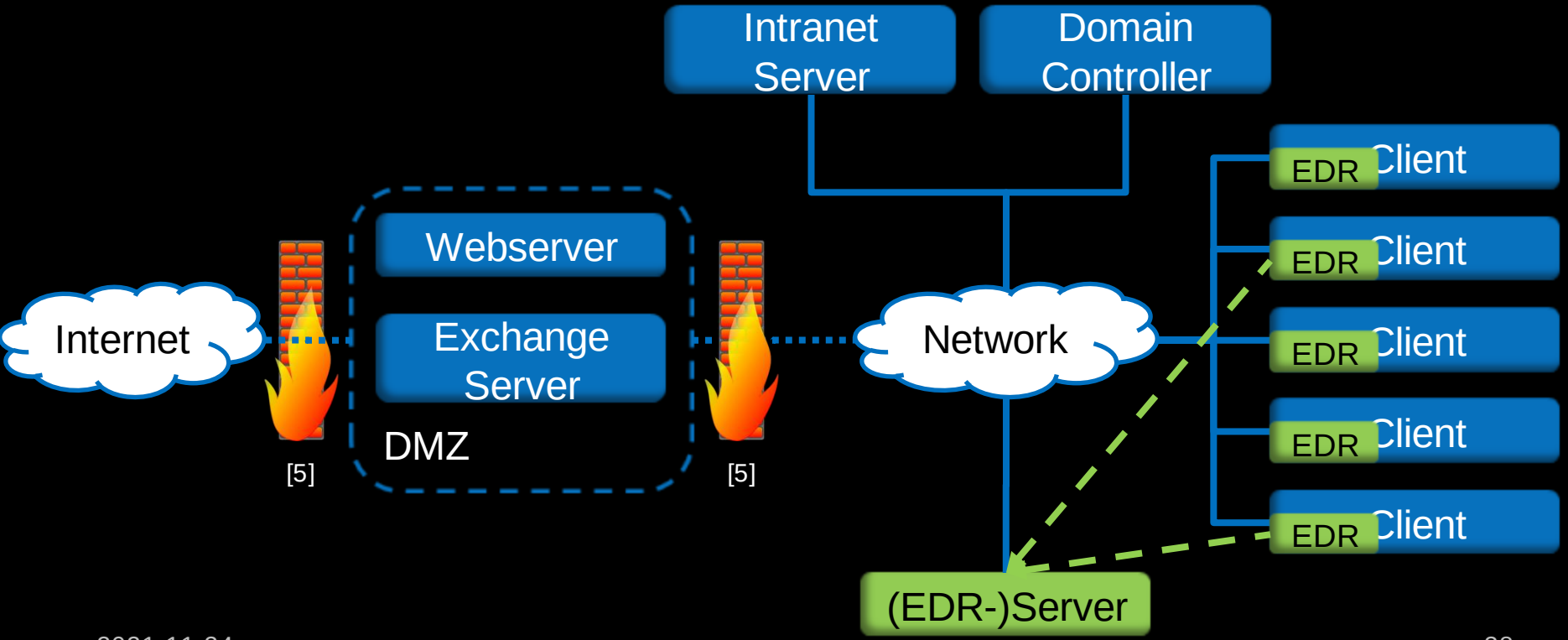
Endpoint Detection & Response (EDR)



Endpoint Detection & Response (EDR)



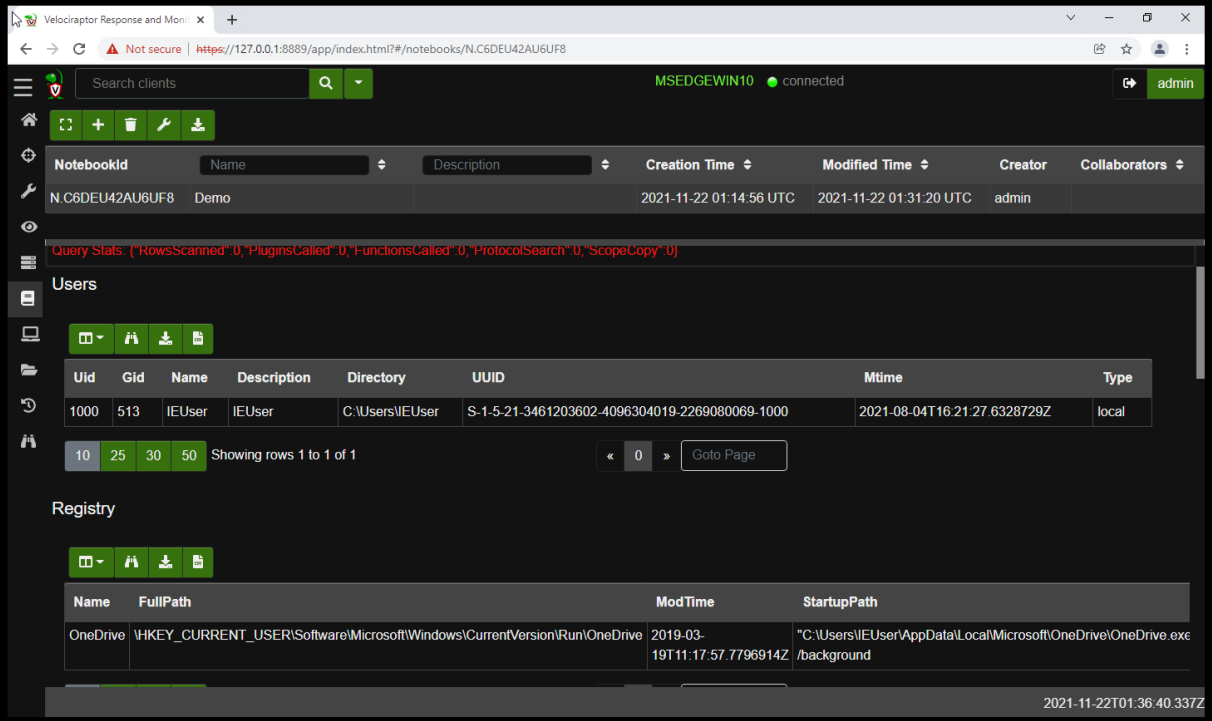
Endpoint Detection & Response (EDR)



velociraptor

- by: Michael Cohen (=volatility)
- open-source EDR
- Client-Server and one-shot collection scripts
- can run file-less (no installation)
- Go binary (runs on all common platforms)

velociraptor Basics



Velociraptor VQL

- Basics

```
SELECT * FROM Artifact.Windows.Sys.Users()
WHERE Name =~ "IE"
```

- Registry

```
SELECT Name, FullPath, ModTime, Data.value AS StartupPath
FROM glob(globs='''/HKEY_CURRENT_USER/**/Run/*''', accessor="registry")
LIMIT 5
```

- Combined Queries

```
SELECT * FROM foreach(
    row={
        SELECT Directory FROM Artifact.Windows.Sys.Users()
    },
    query={
        SELECT FullPath, Mtime FROM glob(globs=Directory+"/*", accessor="ntfs")
    }
)
```

Velociraptor Live-RAM Analysis

```
SELECT Pid, Name,
       format(format="%0#x", args=PebBaseAddress) AS PebBaseAddress,
       parse_binary(accessor="process",
                    filename=format(format="/%v", args=Pid),
                    profile=profile,
                    struct="PEB",
                    offset=PebBaseAddress).ProcessParameters.CommandLine.Buffer AS CommandLine
FROM pslist()
WHERE Name =~ "velociraptor" | Name =~ "Chrome"
```

Explanation:

- „FROM pslist WHERE …“ → get all running tasks named „velociraptor“ or „chrome“
- „parse_binary“ → parse their binary from RAM
- PEB = process environment block (OS structure)
- ProcessParameters.Commandline → get commandline

Velociraptor Live-RAM Analysis

Pid	Name	PebBaseAddress	CommandLine
6600	velociraptor-v0.6.2-windows-amd64.exe	0xa4c7f79000	"C:\Users\IEUser\Downloads\velociraptor-v0.6.2-windows-amd64.exe" gui
6684	chrome.exe	0x68999df000	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument
6748	chrome.exe	0xc9d6b51000	"C:\Program Files\Google\Chrome\Application\chrome.exe" --type=crashpad-h url=https://clients2.google.com/cr/report --annotation=channel= --annotation=p
6880	chrome.exe	0x32a4efd000	"C:\Program Files\Google\Chrome\Application\chrome.exe" --type=gpu-proces preferences=UAAAAAAAAADgAAAYAAAAAAAAAAAAAAAAAABgAIAAAAAw --mojo-platform-channel-handle=1716 /prefetch:2
6896	chrome.exe	0xdcacb7a000	"C:\Program Files\Google\Chrome\Application\chrome.exe" --type=utility --utilit
6968	chrome.exe	0x646bfdf000	"C:\Program Files\Google\Chrome\Application\chrome.exe" --type=utility --utilit

velociraptor Demo

Live-Demo

Up for an adventure in Europe?

CyberSecurity Jobs

Masther thesis in Digital Forensics & Incident Response / student positions:

- Contact me: michael.denzel@airbus.com

Cyber Security Incident Responder (d/f/m):

- https://ag.wd3.myworkdayjobs.com/en-US/Airbus/job/Mnchen-Area/Cyber-Security-Incident-Responder--d-f-m-_JR10038421-1

Cyber Security Internship Penetration Tester (d/f/m):

- https://ag.wd3.myworkdayjobs.com/en-US/Airbus/job/Mnchen-Area/Praktikum-im-Bereich-Penetration-Testing_JR10094534

All other Airbus Cyber Security positions:

- https://www.airbus.com/careers/search-and-apply/search-for-vacancies.html?filters=filter_1_2&textsearch=Cyber

Contact our Cyber Security Recruiter!

- Cornelia Seebauer - cornelia.seebauer@airbus.com
- https://www.linkedin.com/in/cornelia-seebauer-3159b0154

Contact (for research questions)

Dr. Michael Denzel
md.research@mailbox.org

Sources

[1] https://www.blackhat.com/presentations/bh-europe-06/bh-eu-06-Heasman.pdf

[2] https://github.com/mdenzel/ACPI-rootkit-scan

[3] https://github.com/volatilityfoundation/volatility/wiki/Command-Reference#pslist

[4] https://github.com/volatilityfoundation/volatility/wiki/Command-Reference-Mal#ldrmodules

[5] https://pixabay.com/vectors/firewall-fire-wall-computer-153179/

[6] https://unsplash.com/photos/1SAnrIxw5OY

[7] https://unsplash.com/photos/LctoBGY6cR8